Diffusion Models

Learn to remove controlled noise at different corruption levels, then generate by repeatedly denoising an initial noise sample. The schedule, denoiser, and sampling rule jointly define the result.

Original study notes based on David Foster, Second edition, 2023. Book pp. 205–234. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.

Clean image + noise level → Corrupt with known noise → Predict noise → Learn denoising → Iterate from random noise

Detailed Notes

Denoising models, score models, and flower data

Source: book pp. 205–208 / PDF p. 233

Diffusion replaces one-shot generation with a series of smaller denoising problems. A forward process makes data progressively less informative; a learned reverse process restores structure. The flower-image project illustrates image normalization, resizing, batching, and realistic sampling.

Denoising diffusion and score-based models are closely connected through the gradient of log density at different noise levels. Their relationship does not mean every implementation uses the same objective or sampler. Separate the mathematical probabilistic formulation from the simplified continuous-time denoiser used in the project.

Forward diffusion and reparameterization

Source: book pp. 209–210 / PDF p. 237

Mix a clean image with Gaussian noise using time-dependent signal and noise rates. The corrupted sample can be constructed directly at an arbitrary time, without simulating every earlier step.

> $x_t = s(t)x_0 + n(t)\varepsilon,\ \varepsilon \sim N(0,I)$
>
> $s(t)^2 + n(t)^2 = 1$
>
> Discrete notation: $s(t)=\sqrt{\bar{\alpha}_t},\ n(t)=\sqrt{(1-\bar{\alpha}_t)}$

Rates are amplitudes, not variances. Confusing square roots with variances changes the corruption distribution. Data normalization should align with the assumed signal/noise scales. A random noise level per training example efficiently covers the denoising problem.

Diffusion schedules

Source: book pp. 211–213 / PDF p. 239

A schedule controls how information disappears over time. Linear schedules and cosine schedules allocate difficulty differently; an offset cosine keeps endpoint rates away from numerically dangerous extremes.

> $s(t)=\cos(\text{angle}(t)),\ n(t)=\sin(\text{angle}(t))$

Inspect signal rate, noise rate, and signal-to-noise ratio throughout the interval. Very small signal rates make reconstructing the clean image numerically sensitive. The schedule used in training and the interpretation of time during sampling must agree. More noise is a learning condition, not simply arbitrary data augmentation.

Reverse diffusion and noise prediction

Source: book pp. 214–216 / PDF p. 242

The denoiser predicts the noise component conditioned on the corrupted image and noise level. Estimate the clean image, then combine that estimate with predicted noise at a slightly less noisy time.

> $\hat{x}_0 = (x_t - n(t)\hat{\varepsilon}_\theta(x_t,t)) / s(t)$

Do not confuse this deterministic rate-based update with every stochastic DDPM reverse transition. The sampler specifies how uncertainty/noise is introduced between steps. Finite-step error compounds, and dividing by near-zero $s(t)$ is unstable. Preserve the final denoised image rather than returning an arbitrary intermediate noisy state.

The U-Net denoiser

Source: book pp. 217–223 / PDF p. 245

A U-Net downsamples spatial features, processes a compressed representation, then upsamples while combining same-resolution skip features. These connections preserve detail and support both global structure and local corrections.

Encode diffusion time/noise variance, often with sinusoidal features, and inject it into the network so identical pixel values at different noise levels are interpreted differently. Residual blocks ease optimization. Track feature-map shapes across skip concatenations, and match the output channels to the predicted noise. Denoising is a conditioned regression problem, not an image-classification head.

Training, EMA, and reverse sampling

Source: book pp. 224–227 / PDF p. 252

For each batch, sample clean images, Gaussian noise, and times; form corrupted images; predict noise; minimize the chosen denoising error. Reconstruction error can be monitored separately. Match the exact loss/reduction to the implementation rather than mixing incompatible DDPM or tutorial conventions.

Exponential moving averages (EMA) of weights can stabilize sampling. Start generation from Gaussian noise, step through decreasing times using the same schedule semantics, then undo normalization. During inference use inference-mode normalization/weights consistently. Record schedule, step count, seed, and whether EMA weights were used.

Analysis and sampling trade-offs

Source: book pp. 228–234 / PDF p. 256

Compare samples at different reverse-step counts and inspect intermediate denoising trajectories. More steps can improve approximation at higher latency, but sampler design matters too. A low denoising loss alone cannot establish realistic flower diversity or absence of memorization.

Check consistency across seeds, generated diversity, training-data overlap, and failure cases such as implausible geometry. Diffusion is computationally intensive at sample time; latent diffusion and later architectural refinements address part of that cost. Learned denoising is powerful, but output validity still needs domain-specific assessment.

Revision Check

1. Why do the signal and noise rate squares sum to one?
2. How can training sample arbitrary noise times directly?
3. Why is time conditioning essential?
4. What happens near zero signal rate?
5. How do sampling steps and EMA affect evaluation?

A diffusion checkpoint is not the whole generator: data normalization, schedule, time encoding, and reverse sampler must remain compatible.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.